

ACTIVITY 3 — FACILITATOR GUIDE**Analyze — Find the Root Cause**

TYPE	DURATION	MAPS TO	TOPIC
Hands-on Lab	30 minutes	ANALYZE	Analyze — Find the Cause

Purpose of this activity

By the end of this activity learners will be able to identify likely causes using 5 Whys and Fishbone analysis (A3, K2).

What learners produce

A completed 5 Whys chain ending in an actionable cause, a Fishbone diagram with causes sorted into the five categories, and a shortlist of two or three root causes each backed by evidence.

Materials required

- Learner Worksheet (one per learner)
- Checklist (one per group)
- Flipchart or A3 sheet and markers
- The course slides for the scenario and framework
- Internet access where the activity calls for live research

Set-up

Groups of 3–5. Distribute one worksheet per learner and one checklist per group, and make sure every group can open this activity's data/ folder. All 5 activities run on the same continuous BrewBean Cafe scenario and follow DMAIC in order, so remind learners where this activity sits in that story — and that its output feeds the next one.

Scenario / briefing material

The team now knows WHERE the delay happens. This activity asks WHY it happens.

At the last team meeting two theories were argued loudly. The owner still wants to buy a second machine. The shift supervisor wants to add a staff member to the morning rota. Both are solutions to a symptom nobody has yet explained: the queue is long, so add capacity. Neither theory has been tested against the observation data.

Most improvement teams jump straight from a symptom to a solution and end up fixing the wrong thing at considerable cost. Your job is to drill past the symptom to a cause the team can actually act on — and to be able to show the evidence for it.

How to run it — step by step

#	Instruction to learners	Facilitator note
1	State the symptom. Write the	Watch for chains that stop at 'the

	symptom from your Lab 2 baseline at the top of templates/5-whys-template.csv — for example 'customers wait an average of 8.9 minutes against a 5-minute promise'. Start from the measured symptom, not from a theory about the cause.	barista is slow'. That blames a person. Ask 'why is the barista slow?' and keep going.
2	Drill down with 5 Whys. Ask 'why does that happen?' down the template, each answer becoming the next question. Stop when you reach a PROCESS the team can act on. Two rules: never stop at a person ('the barista is slow' is not a root cause — ask why), and do not stop early just because an answer sounds plausible.	Some tables stop after two whys. The test is not exactly five levels — it is reaching something the team can act on.
3	Set up the Fishbone. Open templates/fishbone-template.csv and write the problem in the head of the fish. The 5 Whys drills DOWN one chain; the Fishbone spreads WIDE so no whole area of cause is missed. Used together they are stronger than either alone.	If a cause appears on the fishbone with no evidence anywhere, point at the cause-evidence sheet. Opinion is fine in brainstorming, not in the shortlist.
4	Populate the five bones. Brainstorm possible causes onto Manpower, Method, Machine, Material and Measurement. Then check them against data/cause-evidence-sheet.csv to see which ones the data actually supports — some entries there are pure opinion, and telling them apart is the point.	Most rooms label everything 'common cause'. Use the grinder breakdown on the evidence sheet as the anchor example of a special cause.
5	Read the Pareto. Open data/delay-reason-pareto-data.csv — the delay reasons from your Lab 2 log, counted and ranked. Read the Cumulative % column and state which few causes account for most of the problem. Remember the Pareto ranks FREQUENCY, which is usually but not always the same as importance.	Compare two tables' 5 Whys chains side by side at the debrief. They usually reach different root causes from the same symptom — which is exactly why the evidence column decides. Flag that Case Study Q2 asks for this chain.
6	Separate common from special cause. Using the last column of the cause-evidence sheet, decide for each cause whether it is common cause (built into the process, happens every day) or special cause (a specific one-off event). Common causes are fixed by changing the process; a team that chases special causes is always firefighting.	
7	Shortlist on evidence. In templates/root-cause-shortlist-template.csv, shortlist the two or three causes best supported by	

	evidence — not by opinion, and not by who argued loudest. For each, record the evidence and where it came from. Three is the maximum a White Belt team should carry into Improve.	
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Suggested timing

Minutes	Phase
9 min	Build the 5 Whys chain down to an actionable cause
8 min	Spread the causes across the Fishbone categories
5 min	Read the Pareto and name the vital few
4 min	Separate common cause from special cause
4 min	Shortlist the causes backed by evidence

Total: 30 minutes

Discussion & decision prompts

1. Compare your 5 Whys chain with another group's. You started from the same symptom — did you reach the same root cause, and what does that tell you?
2. Which causes on your Fishbone have real evidence behind them, and which are assumptions the team simply believes?
3. The Pareto ranks causes by frequency. When might the most frequent cause NOT be the most important one?
4. Why does it matter whether a cause is common cause or special cause?

Debrief

Bring the class back together and draw out the learning:

- Ask two or three groups to show what they produced, not just what they concluded.
- Ask what surprised them — the gap between what they assumed and what they found.
- Link the activity explicitly back to DMAIC · ANALYZE — Analyze — Find the Cause.
- Remind learners this activity rehearses what the Case Study assessment will require.

Reflection questions:

- Did your chain stop at a person at any point? What happened when you asked 'why' again?
- Which of the owner's or supervisor's two theories survives contact with the data?
- What would it have cost the cafe to act on the loudest opinion instead of the evidence?

Facilitator notes — what to watch for

- Watch for chains that stop at 'the barista is slow'. That blames a person. Ask 'why is the barista slow?' and keep going.
- Some tables stop after two whys. The test is not exactly five levels — it is reaching something the team can act on.
- If a cause appears on the fishbone with no evidence anywhere, point at the cause-evidence sheet. Opinion is fine in brainstorming, not in the shortlist.
- Most rooms label everything 'common cause'. Use the grinder breakdown on the evidence sheet as the anchor example of a special cause.

- Compare two tables' 5 Whys chains side by side at the debrief. They usually reach different root causes from the same symptom — which is exactly why the evidence column decides. Flag that Case Study Q2 asks for this chain.

Success criterion

Every group can show a 5 Whys chain ending in an actionable process cause, a Fishbone with causes sorted into the five categories, and two or three shortlisted causes each backed by a stated observation.

Assessment link

This activity builds ANALYZE — Analyze — Find the Cause. All 5 activities are in-class activities assessed indirectly through the open-book Case Study (CS) component, which reuses the same techniques on the same continuous BrewBean Cafe scenario. Knowledge is assessed in the Written Assessment (WA-SAQ). Nothing is assessed that is not practised here or taught in the slides.